

Comparative Evaluation: CrysMMNet vs. Causality-LLM Paper for UAI

Executive Summary

CrysMMNet (UAI 2023 - Accepted): Strong empirical work in materials science with clear novelty **Your Causality Paper (UAI 2026 - Target):** Solid methodological contribution with room for improvement

Overall Assessment: Your causality paper is competitive for UAI 2026 but needs strengthening in theoretical depth and experimental breadth. Current quality: **7/10** for UAI standards.

Detailed Comparison Across UAI Evaluation Criteria

1. NOVELTY & CONTRIBUTION

CrysMMNet

- **Novel Contribution:** First multimodal (graph + text) approach for crystal property prediction
- **Dataset Contribution:** Curated textual descriptions for MP and JARVIS datasets
- **Methodological Innovation:** Fusion of structural and textual representations
- **Impact:** Clear advancement over single-modality SOTA (4-15% improvements)

Score: 9/10

Your Causality Paper

- **Novel Contribution:** Variance-aware evaluation framework for assessing LLM meta-knowledge
- **Methodological Innovation:** Bootstrap-based confidence intervals for stochastic algorithm evaluation
- **Gap Addressed:** Prior work uses single-run comparisons (methodologically flawed)
- **Weaknesses:**
 - Concept of "meta-knowledge" not deeply theorized
 - Limited to 3 classical algorithms (no modern methods like NOTEARS, GOLEM)
 - Missing: Why is this important beyond empirical curiosity?

Score: 7/10

Gap Analysis: Your paper needs stronger motivation for *why* understanding LLM meta-knowledge matters practically. CrysMMNet clearly motivates the multimodal approach (global vs. local structure). You should add:

- Concrete use cases where wrong LLM predictions about algorithms would cause harm
- Connection to hybrid human-AI workflows in scientific discovery
- Framework for when to trust vs. verify LLM algorithm recommendations

2. TECHNICAL RIGOR

CrysMMNet

- **Experimental Design:**
 - 10 properties across 2 datasets
 - Consistent train/val/test splits
 - Ablation studies (text-only, different GNN backbones, global vs. local)
- **Statistical Analysis:** Standard ML evaluation (MAE, confidence from splits)
- **Reproducibility:** Code + dataset released
- **Weaknesses:**
 - No theoretical analysis of why fusion helps
 - Limited discussion of when method fails

Score: 8/10

Your Causality Paper

- **Experimental Design:**
 - 100 runs per algorithm-dataset pair (excellent)
 - Bootstrap CIs with 10,000 iterations (rigorous)
 - 9 datasets spanning real-world, benchmarks, synthetic
 - Bonferroni correction for multiple comparisons
- **Statistical Analysis:** Strong (t-tests, correlation, overlap metrics)
- **Reproducibility:** Missing code/data availability statement
- **Strengths:**
 - Variance-aware framework is methodologically sound
 - Non-parametric bootstrap avoids distributional assumptions
- **Weaknesses:**
 - Only 2 LLMs tested (need Claude, Gemini for robustness)
 - Single prompt per configuration (acknowledged, but limits conclusions)
 - No theoretical framework for *why* LLMs would have meta-knowledge
 - Missing: Sensitivity analysis on prompt variations

Score: 7.5/10

Critical Missing Element: You need the prompt variation experiment you mentioned. UAI reviewers will ask: "How sensitive are these results to prompt phrasing?" Without this, your conclusions are fragile.

3. EXPERIMENTAL BREADTH & DEPTH

CrysMMNet

- **Breadth:** 2 datasets, 10 properties, 7 baseline comparisons
- **Depth:**
 - Ablations isolate contribution of each component

- Attention visualization provides interpretability
 - Limited data regime tested (shows robustness)
- **Completeness:** Addresses obvious questions (Is text alone enough? Does fusion help all GNNs?)

Score: 8.5/10

Your Causality Paper

- **Breadth:**
 - 3 algorithms (LiNGAM, PC, FCI)
 - 9 datasets (good coverage)
 - 4 metrics (Precision, Recall, F1, SHD)
 - 2 LLMs
- **Depth:**
 - Strong variance quantification
 - Good identification of systematic biases ($2.3\times$ SHD underestimation)
 - Context-dependence analysis (simple vs. complex graphs)
- **Gaps:**
 - **Critical:** Only 2 LLMs (need at least 4-5 for UAI)
 - **Critical:** Missing prompt variation analysis
 - Modern causal discovery methods absent (NOTEARS, DAG-GNN, GOLEM)
 - No interventional data experiments (PsiFCI mentioned but not tested)
 - Limited real-world datasets (only Titanic, Sachs)

Score: 6.5/10

Major Gap: UAI expects comprehensive coverage. Your dataset selection is good but algorithm coverage is dated (2011, 1991, 2000). Adding even 1-2 modern gradient-based methods would significantly strengthen the paper.

4. WRITING QUALITY & CLARITY

CrysMMNet

- Clear motivation and problem setup
- Well-structured with logical flow
- Good use of figures (especially Figure 1 showing text content)
- Tables are comprehensive and easy to parse
- Minor issues: Some repetition in ablation discussion

Score: 8/10

Your Causality Paper

- **Strengths:**
 - Clear articulation of the variance-aware evaluation problem
 - Good organization (Background → Methodology → Results → Discussion)
 - Tables are well-formatted and informative

- Statistical rigor is transparent
- **Weaknesses:**
 - **Abstract is too long** (standard limit is ~150-200 words, yours is ~200+)
 - Section 1.1 ("Challenge of Evaluating LLM Meta-Knowledge") could be tighter
 - Results section is dense—needs more interpretive guidance
 - **Missing:** Related work on LLM evaluation methodologies (beyond causal reasoning)
 - Discussion of limitations is good but comes too late (should preview in intro)

Score: 7/10

Specific Improvements Needed:

- Tighten abstract (remove "Keywords" line—UAI uses standardized format)
- Add 2-3 sentences in intro previewing main findings
- Results: Add interpretive paragraphs between result tables
- Related work: Add subsection on "LLM Evaluation Methodologies" citing variance-aware evaluation in other domains

5. IMPACT & SIGNIFICANCE

CrysMMNet

- **Domain Impact:** High—materials discovery is economically important
- **ML Impact:** Moderate—shows multimodal learning works in scientific domains
- **Citations:** Likely high (materials + ML communities)
- **Practical Value:** Clear (faster materials screening)

Score: 8/10

Your Causality Paper

- **Domain Impact:**
 - Causal discovery: Moderate (clarifies LLM limitations)
 - LLM evaluation: High (introduces rigorous framework)
- **Practical Value:**
 - Informs when to trust LLM recommendations about algorithms
 - Provides template for evaluating LLM knowledge of stochastic systems
- **Broader Applicability:** Framework extends to optimization, simulation, other stochastic algorithms
- **Weaknesses:**
 - Findings are somewhat negative (LLMs don't work well) → less exciting than positive results
 - Unclear if anyone is *currently* using LLMs this way (motivates as preventing a problem vs. solving one)

Score: 7/10

Impact Concern: UAI reviewers may ask "Who is the audience?" You position this for:

1. Scientists considering LLM-assisted causal discovery

2. LLM developers improving algorithmic reasoning
3. Methodologists evaluating LLM capabilities

This is good breadth, but you need to strengthen the "preventing harm" narrative. Add examples of cases where wrong LLM predictions about algorithm performance led to poor decisions (even if hypothetical/synthetic).

6. METHODOLOGICAL SOUNDNESS

CrysMMNet

- Follows established ML pipeline
- Appropriate baselines (all recent SOTA)
- Fair comparisons (uses published results to avoid implementation bias)
- Ablations isolate contributions
- **Minor Issue:** Freezing MatSciBERT weights is pragmatic but limits potential gains

Score: 8/10

Your Causality Paper

- Bootstrap methodology: Sound
- Overlap metrics: Well-designed (binary, percentage, containment)
- Statistical testing: Appropriate (t-tests with correction)
- Experimental controls: Good (synthetic data, known ground truth)
- **Strengths:**
 - Non-parametric approach avoids assumptions
 - Multiple metrics provide triangulation
 - Correlation analysis identifies patterns
- **Weaknesses:**
 - **Prompt engineering:** Single prompt is a major limitation
 - **LLM sampling:** No temperature variation or multiple samples
 - **Ground truth:** Some benchmarks have known DAGs, others use "domain knowledge" (Titanic)—this asymmetry is not discussed
 - **Algorithm parameters:** Hyperparameter ranges are described but not justified (why $\alpha \in [0.01, 0.10]$?)

Score: 7/10

Critical for UAI: The single-prompt limitation is severe. Reviewers will likely request:

- At least 3 prompt variations (declarative, interrogative, few-shot)
- Temperature sweep (0.0, 0.5, 0.9) with multiple samples
- Sensitivity analysis showing results are robust to formulation

Your acknowledgment in "Future Work" isn't sufficient—this needs to be in the main results.

SPECIFIC WEAKNESSES THAT MUST BE ADDRESSED

Critical (Will likely cause rejection if not fixed):

1. **Prompt Variation Experiment:** You MUST run the 3 prompt variants you mentioned. This is non-negotiable for UAI.
2. **LLM Coverage:** Add at least 2 more LLMs (Claude 3.5 Sonnet, Gemini 1.5 Pro). With only GPT-5 and DeepSeek, conclusions about "LLMs" in general are weak.
3. **Algorithm Coverage:** Add at least 1 modern method (NOTEARS is easiest—differentiable, well-known). Covering only 20-year-old algorithms makes the work feel dated.

Major (Will strengthen significantly):

4. **Theoretical Framework:** Add a section explaining *why* LLMs might encode meta-knowledge. Is it from:
 - Memorization of published benchmarks?
 - Implicit reasoning about algorithm structure?
 - Pattern matching from algorithm descriptions?

Propose hypotheses and design experiments to distinguish them.

5. **Real-World Datasets:** Add 2-3 more real datasets from different domains (finance, epidemiology, ecology). Titanic and Sachs are overused.
6. **Failure Mode Analysis:** Deepen analysis of *why* LLMs fail on complex graphs. Is it:
 - Combinatorial explosion (can't reason about large search spaces)?
 - Training data bias (more papers on small graphs)?
 - Lack of mechanistic understanding?

Minor (Would improve polish):

7. **Reproducibility:** Add code/data availability statement
8. **Figure Quality:** Add visualizations:
 - Overlap % vs. graph complexity (scatter plot)
 - LLM prediction ranges vs. algorithmic CIs (error bars)
 - Heatmap of overlap across algorithm-dataset pairs
9. **Related Work:** Expand section 2.2 to include:
 - LLM evaluation methodologies (beyond causal reasoning)
 - Variance-aware evaluation in other domains
 - Meta-learning and algorithm selection

COMPARISON TO UAI STANDARDS

What UAI Accepts:

- Novel problem formulations with rigorous solutions
- Strong empirical work with thorough evaluation
- Methodological contributions applicable beyond one domain

- Clear practical or theoretical impact

CrysMMNet (2023 Acceptance):

- ✓ Novel problem (multimodal materials)
- ✓ Strong empirics (10 properties, consistent improvements)
- ✓ Methodological contribution (fusion framework)
- ✓ Practical impact (faster materials discovery)
- ✓ Reproducible (code + data released)

Your Paper (Current State):

- ✓ Novel problem (variance-aware LLM evaluation)
- Δ Empirics: Strong methodology, but coverage gaps (2 LLMs, 3 algorithms, 1 prompt)
- ✓ Methodological contribution (bootstrap-based framework)
- Δ Impact: Valuable but somewhat negative (LLMs don't work well)
- ✗ Reproducibility: No availability statement

Verdict: Current version is borderline. With critical fixes (prompt variants, more LLMs), likely accept. Without them, likely reject.

ACTIONABLE RECOMMENDATIONS (Priority Order)

Must Do (Before Submission):

- 1. Run Prompt Variation Study:**
 - 3 formulations $\times$ 2 LLMs $\times$ 27 configs = 162 queries (doable)
 - Report mean $\pm$ std across prompts
 - Show results are robust ($\pm 10\%$ variance) or sensitive (needs discussion)
- 2. Add 2-3 More LLMs:**
 - Claude 3.5 Sonnet (Anthropic)
 - Gemini 1.5 Pro (Google)
 - Llama 3.1 70B (open-source baseline)
 - This makes "LLM" conclusions generalizable
- 3. Add 1-2 Modern Algorithms:**
 - NOTEARS (differentiable, ICLR 2018)
 - DAG-GNN (NeurIPS 2019) or GOLEM (NeurIPS 2020)
 - Even if results are similar, shows comprehensive coverage
- 4. Strengthen Motivation:**
 - Add paragraph in intro: "Why does meta-knowledge matter?"
 - Include examples of potential harm from wrong predictions
 - Connect to hybrid human-AI scientific workflows
- 5. Add Code/Data Availability:**
 - "Code and experimental data will be released upon acceptance"
 - Or release anonymously during review if allowed

Should Do (Will Strengthen):

6. **Expand Real-World Datasets:**
 - Add Alarm network (medical)
 - Add financial/economic causal graph
 - Strengthens generalizability claims
7. **Add Theoretical Section:**
 - Formalize "meta-knowledge" (what does it mean for an LLM to "know" algorithm behavior?)
 - Propose mechanisms (memorization vs. reasoning)
 - Design experiments to test (e.g., novel algorithm LLMs haven't seen)
8. **Improve Figures:**
 - Figure showing overlap vs. complexity (with trend line)
 - Error bar plots comparing LLM ranges to algorithmic CIs
 - Visual summary of systematic biases
9. **Expand Related Work:**
 - LLM evaluation methodologies (perplexity, MMLU, etc.)
 - Variance-aware evaluation in ML (bootstrapping, cross-validation theory)
 - Algorithm selection and meta-learning

Nice to Have (Polish):

10. **Attention Analysis:**
 - Probe LLM attention on prompt components (dataset description, algorithm assumptions)
 - Show what LLMs attend to when predicting performance
 - Explains mechanism behind predictions
11. **Case Studies:**
 - Deep dive into 2-3 algorithm-dataset pairs
 - Show detailed LLM reasoning (chain-of-thought)
 - Analyze where reasoning breaks down
12. **User Study:**
 - Ask domain experts (causality researchers) to predict algorithm performance
 - Compare human vs. LLM vs. algorithmic ground truth
 - Shows LLMs are not worse than humans (or are worse)

ESTIMATED SCORES

CrysMMNet (UAI 2023):

- Novelty: 9/10
- Technical Rigor: 8/10
- Experiments: 8.5/10
- Writing: 8/10
- Impact: 8/10
- **Overall: 8.3/10** (Strong Accept)

Your Paper (Current):

- Novelty: 7/10

- Technical Rigor: 7.5/10
- Experiments: 6.5/10 (due to coverage gaps)
- Writing: 7/10
- Impact: 7/10
- **Overall: 7/10** (Borderline—could go either way)

Your Paper (With Critical Fixes):

- Novelty: 7.5/10 (stronger with more LLMs/algorithms)
 - Technical Rigor: 8.5/10 (prompt variants add robustness)
 - Experiments: 8/10 (comprehensive coverage)
 - Writing: 7.5/10 (minor improvements)
 - Impact: 7.5/10 (stronger motivation)
 - **Projected: 7.8/10** (Accept)
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TIMELINE RECOMMENDATIONS

Assuming UAI 2026 submission deadline is ~February 2026:

Immediate (Next 2 Weeks):

- Run prompt variation study (3 variants $\times$ existing configs)
- Add Claude, Gemini, Llama queries (rerun existing prompts)
- This gives you 5 LLMs $\times$ 3 prompts = 15 $\times$ current data

Short-Term (1 Month):

- Implement NOTEARS and rerun experiments
- Add 2 more real-world datasets
- Write theoretical section on meta-knowledge

Medium-Term (2 Months):

- Create figures/visualizations
- Expand related work
- Write up new results
- Internal review with collaborators

Final (1 Month Before Deadline):

- Polish writing
 - Ensure reproducibility
 - Prepare supplementary materials
 - Proofread for clarity
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CONCLUSION

Your causality paper has a solid foundation but needs strengthening to meet UAI standards. The variance-aware evaluation framework is genuinely novel and methodologically sound—this is your core contribution.

Key Differentiator from CrysMMNet:

- CrysMMNet solves a clear problem (better materials prediction) with a novel approach (multimodal fusion)
- Your paper proposes a new evaluation methodology (variance-aware) for an emerging question (do LLMs understand algorithms?)

Both are valuable, but yours requires more work to convince reviewers the question is important and the answer is comprehensive.

Bottom Line:

- Without fixes: **40% chance of acceptance** (reviewers will cite incomplete coverage)
- With critical fixes: **70% chance of acceptance** (solid methodological contribution)
- With all recommendations: **85% chance of acceptance** (comprehensive study with clear impact)

You have a publishable idea—now you need to execute thoroughly. The prompt variation and additional LLMs are non-negotiable. The modern algorithms and theoretical framework would elevate it from "solid work" to "significant contribution."

Good luck with your submission!